Geometry and Spaces of Forms: Shape deformation analysis from the optimal control viewpoint

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Problem

Definition: Problem

Deformations, and diffeormophisms in particular, have played a tremendous role in the field of statistical shape analysis, as a proxy to measure and interpret differences between similar objects but with different shapes.

Presentation of the paper: "Shape deformation analysis from the optimal control viewpoint" [1], published in the Journal de Mathématiques Pures et Appliquées in 2015, which presents a fairly general framework for form spaces.

Deformations

Definition: Deformation of class $\mathcal{C}^l$ [1]

With $v \in L^1([0,1]; \mathcal{C}_0^l(\mathbb{R}^d, \mathbb{R}^d))$, the flow:

$$\forall t \in [0,1],\ \dot{\varphi}(t) = v(t) \circ \varphi(t), \quad \varphi(0) = \mathrm{Id}_{\mathbb{R}^d} \tag{1}$$

Property: The set of deformations [1]

$$\left\{ \varphi \in W^{1,1}([0,1]; \mathrm{Diff}_0^l(\mathbb{R}^d)) \mid \varphi(0) = \mathrm{Id}_{\mathbb{R}^d} \right\}$$

Elements of proof: We can use Grönwall inequality and Faà di Bruno's formula. We can recover v using $v(t) = \dot{\varphi}(t) \circ \varphi^{-1}(t)$

Reproducing Kernel Vector Space

Definition: **Reproducing Kernel Vector Space of vector fields of class $\mathcal{C}^l$ [1]**

Hilbert space $(V, \langle \cdot, \cdot \rangle_V)$ of vector fields on $\mathbb{R}^d$, such that $V \subset C_0^l(\mathbb{R}^d, \mathbb{R}^d)$ with continuous inclusion (i.e. $\exists C > 0$, $\|v\|_l \leq C\|v\|_V$).

Action and infinitesimal action

Definition: Action [1]

Map $(\varphi, q) \in \mathrm{Diff}_0^l(\mathbb{R}^d) \times M \mapsto \varphi \cdot q \in M$ is such that $\mathrm{Id}_{\mathbb{R}^d} \cdot q = q$ and $\varphi \cdot (\psi \cdot q) = (\varphi \circ \psi) \cdot q$ for all $\psi, \varphi \in \mathrm{Diff}_0^l(\mathbb{R}^d)$ and $q \in M$.

Definition: Infinitesimal action [1]

Map ξ_q of $\mathcal{C}_0^l(\mathbb{R}^d, \mathbb{R}^d)$ is defined as the differential of $\varphi \mapsto \varphi \cdot q$ at identity.

Problem 1

Definition: **Problem 1 [1]**

Minimizing the functional:

$$\boxed{J_1(q,v) = \frac{1}{2}\int_0^1 \|v(t)\|_V^2\, dt + g(q(1))} \tag{2}$$

Over all $q(\cdot) \in W^{1,1}_{q_0}([0,1]; M)$, $v(\cdot) \in L^2([0,1], V)$ such that $\dot{q}(t) = \xi_{q(t)} v(t)$ and $C_{q(t)} v(t) = 0$ for almost every $t \in [0,1]$.

Theorem 1

Theorem: Theorem 1 [1]

If V is an RKHS of vector fields of class C^{l+1} on $\mathbb{R}^d$, with $q \mapsto C_q$ continuous, and if g is bounded below and lower semi-continuous. If q_0 has compact support, then Problem 1 has at least one solution.

Elements of proof: The proof is based on James theorem, Arzela–Ascoli theorem, diagonal extraction and the properties of lower semi-continuity.

Problem 2

Definition: **Problem 2 [1]**

Minimizing the functional:

$$\boxed{J_2(q,u) = \frac{1}{2}\int_0^1 (u(t), K_{q(t)}u(t))_{X^\star,X}\,dt + g(q(1))} \tag{3}$$

Over $u : [0,1] \to X^\star$ a measurable function and $q(\cdot) \in W^{1,1}_{q_0}([0,1];M)$ such that $\dot{q}(t) = K_{q(t)}u(t)$ and $C_{q(t)}K_V\xi^\star_{q(t)}u(t) = 0$ for almost every $t \in [0,1]$.

Equivalence problem 1 and 2

Property: **Equivalence [1]**

Under some hypothesis, problems 1 and 2 are equivalent.

Moreover, if $(q(\cdot), u(\cdot))$ is an optimal solution of Problem 2, then $(q(\cdot), v(\cdot))$ is an optimal solution of Problem 1, with $v(\cdot) = K_V \xi^\star_{q(\cdot)} u(\cdot)$.

Conversely, if $(q(\cdot), v(\cdot))$ is an optimal solution of Problem 1 then there exists a measurable function $u : [0,1] \to X^\star$ such that $v(\cdot) = K_V \xi^\star_{q(\cdot)} u(\cdot)$ and $(q(\cdot), u(\cdot))$ is an optimal solution of Problem 2.

Elements of proof: Use $v(t) = K_V \xi_{q(t)} u(t)$ for one inequality and the decomposition of $v(t)$ over $\mathrm{Null}(\xi_{q(t)}) \oplus \mathrm{Null}(\xi_{q(t)})^\perp$ for the other inequality, using $\mathrm{Null}(\xi_{q(t)})^\perp = \mathrm{Range}(K_V \xi^\star_{q(t)})$

Examples: Landmarks

Definition: The space of landmarks [1]

Let $X = (\mathbb{R}^d)^n$ and $M = \mathrm{Lmk}_d(n) := \{(x_1, \cdot, x_n) \in (\mathbb{R}^d)^n \mid x_i \neq x_j \text{ if } i \neq j\}$.
With $q = (x_1, \cdot, x_n)$, $\varphi \cdot q = (\varphi(x_1), \cdots, \varphi(x_n))$.
Then :

$$v(t) = \sum_{i=1}^{n} K(x_j(t), \cdot) u_j(t), \quad K_q = [K(x_i, x_j)]_{i,j} \in \mathcal{M}_{nd}(\mathbb{R})$$

Problem 3

Definition: **Problem 3 [1]**

We consider the optimal control problem of minimizing the functional:

$$\boxed{J(q,v) = \int_0^1 L(q(t), v(t)) + g(q(1))} \tag{4}$$

Over all $q(\cdot) \in H^1([0,1]; M)$ and $v(\cdot) \in L^2([0,1]; V)$ such that $\dot{q}(t) = \xi_{q(t)} v(t)$ and $C_{q(t)} v(t) = 0$.

Theorem 2

Theorem: Theorem 2 [1]

Assume that the linear operator $C_q : V \to Y$ is surjective for every $q \in M$. Let $q(\cdot) \in H^1_{q_0}([0,1];X)$ and $v(\cdot) \in L^2([0,1];V)$ be an optimal solution of Problem 3. Then there exist $p \in H^1([0,1];X^\star)$ and $\lambda \in L^2([0,1];Y^\star)$ such that $p(1) + dgq(1) = 0$ and for almost every $t \in [0,1]$:

$$\begin{cases} \dot{q}(t) = \partial_p H(q(t), p(t), v(t), \lambda(t)) \\ \dot{p}(t) = -\partial_q H(q(t), p(t), v(t), \lambda(t)) \\ \qquad \partial_v H(q(t), p(t), v(t), \lambda(t)) = 0 \end{cases} \tag{5}$$

Deformation and Compound module

Definition: Deformation Module [3]

We define $(M, \mathcal{H}, \xi, \zeta, c)$ as a $\mathcal{C}^k$-*deformation module* of order l with *geometrical descriptors* in M, h *controls* in $\mathcal{H}$, infinitesimal action ξ if:
The *field generator* $\zeta : M \times \mathcal{H} \to \mathcal{C}_0^l(\mathbb{R}^d, \mathbb{R}^d)$ is continuous, linear with respect to h and $\mathcal{C}^k$ with respect to q.
The *cost* c is smooth and a positive quadratic form on $\mathcal{H}$.

Definition: Compound Module [3]

Let $(M_l, \mathcal{H}_l, \xi_l, \zeta_l, c_l)$, $l = 1 \cdots L$, be L $\mathcal{C}^k$-deformation modules of order l. We define the *compound module* of modules these modules $(M, \mathcal{H}, \xi, \zeta, c)$ where $M = \prod_l M_l$, $\mathcal{H} = \prod \mathcal{H}_l$,and :

$$\zeta_q h = \sum_l \zeta_{q_l}^l h_l$$

$$\xi_q v = (\xi_{q_l}^l v_l)_l$$

$$c_q(h) = \sum_l c_{q_l}^l (h_l)$$

Uniform Embedding Condition

Definition: Uniform Embedding Condition: UEC [3]

We say that a deformation module satisfies the *Uniform Embedding Condition* (UEC) if there exists a Hilbert space of vector fields V continuously embedded in $\mathcal{C}_o^{l+k}(\mathbb{R}^d, \mathbb{R}^d)$ and a constant $\gamma > 0$ such that :

$$\forall q \in M, \forall h \in \mathcal{H}, \quad \zeta_q h \in V, \quad \|\zeta_q(h)\|_V^2 \leq \gamma c_q(h) \tag{6}$$

Property: UEC for compound modules [3]

If $(M_l, \mathcal{H}_l, \xi_l, \zeta_l, c_l)$, with $l = 1 \cdots L$, are $\mathcal{C}^k$-deformation modules of order l, then their compound module is also a $\mathcal{C}^k$-deformation module of order l. Furthermore, if they all satisfies UEC, then their compound module also satisfies UEC.

Elements of proof: We define $\sigma : (w_1, \cdots, w_L) \in W = \prod_l V_l \mapsto \sum_l w_l$. Then with $V = \sigma(W)$ equipped with the norm $|v|_V^2 = \inf\{\sum_l |v_l|_{V_l}^2 \mid \sigma((v_l)_l) = v\}$.

Problem 4

Definition: **Problem 4 [3]**

Let $(a, b) \in M^2$. We consider the problem of minimizing the functional:

$$\boxed{J_{a,b}(h) = g(q(1), b) + \int_0^1 c_{q(t)}(h(t))dt} \tag{7}$$

Such that $v(t) = \zeta_{q(t)} h(t)$, and with $q(t) = \varphi^v(t) \cdot a$.

Theorem 4

Theorem: Theorem 4 [3]

With $(M, \mathcal{H}, \zeta, \xi, c)$ a $\mathcal{C}^k$ deformation module of order $\mathcal{C}^l$ satisfying UEC with $k, l \geq 1$. We suppose that g is $\mathcal{C}^1$. If $h \in L^2([0,1], \mathcal{H})$ minimizes functional $J_{a,b}$ then there exists a path $p : t \in [0,1] \to p(t) \in T^\star_{q(t)}M$ such that with :

$$H(q,p,h) = (p, \xi_q \zeta_q h) - \frac{1}{2} c_q(h) \tag{8}$$

The *hamiltonian* of the system and $p(1) + \partial_q g(q(1), b) = 0$:

$$\begin{cases} \dot{q}(t) = \xi_{q(t)} \zeta_{q(t)} h(t) \\ \dot{p}(t) = -\dfrac{\partial H}{\partial q}(q(t), p(t), h(t)) \\ 0 = \dfrac{\partial H}{\partial h}(q(t), p(t), h(t)) \end{cases} \tag{9}$$

Implicit Deformation Modules

Definition: **Implicit Deformation Modules [2]**

Let M be the space of geometrical descriptors, $\mathcal{H}$ be the space of controls and V be a RKHS continuously embedded in $\mathcal{C}_0^l(\mathbb{R}^d)$ with $l \geq 1$. We consider: A *space of constraints* Y which is a vector space of finite dimension.

An *evaluation operator* $S : M \times V \to Y$, linear in v, globally smooth such that for all $q \in M$, $S(q, \cdot)$ is surjective.

A *prescription operator* $A : M \times \mathcal{H} \to Y$, linear in $h \in \mathcal{H}$ and smooth.

And a *relaxing constant* $\nu > 0$.

Implicit Deformation Modules = Deformation Modules

Property: **Implicit Deformation Modules as Deformation Modules [2]**

Let $(M, \mathcal{H}, \xi, \zeta, c)$ be the *implicit deformation module* induced by the constraints (S, A):

The associated field generator is $\zeta_q(h) = K_V S_q^\star (S_q K S_q^\star + \nu I)^{-1} A_q(h)$
The associated cost is $c_q(h) = ((S_q K S_q^\star + \nu I)^{-1} A_q(h) | A_q(h))$

Rmk: The field generator is solution of the following minimisation problem

$$\zeta_q(h) = \underset{v \in V}{\mathrm{Argmin}} \frac{1}{\nu} \|S_q(v) - A_q(h)\|^2 + \|v\|_V^2$$

Thank you for your attention.

[1] Sylvain Arguillere et al. "Shape deformation analysis from the optimal control viewpoint". In: *Journal de mathématiques pures et appliquées* 104.1 (2015), pp. 139–178.

[2] Benjamin Charlier et al. "Structured Deformation Modeling with Implicit Deformation Modules". In: (2026).

[3] Barbara Gris, Stanley Durrleman, and Alain Trouvé. "A sub-riemannian modular framework for diffeomorphism-based analysis of shape ensembles". In: *SIAM Journal on Imaging Sciences* 11.1 (2018), pp. 802–833.